

3D PRINTERS

Costing Under ₹ 100,000



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3D printing, also known as additive manufacturing, is considered to be a disruptive technology that could power the next industrial revolution. It could transform the way designers visualise concepts, the way companies do businesses and, particularly, the way manufacturers make products.

When the design process moves forward technical decisions are iteratively tested to achieve improved performance, lower manufacturing costs, deliver higher quality and more successful product introductions. In pre-production, 3D printing is enabling faster first object production to support marketing and sales functions, and early adopter customers. And in final production processes, 3D printing is enabling higher productivity, increased flexibility, reduced warehouse and other logistics costs, economical customisation, improved quality, reduced product weight and greater efficiency in a growing number of industries.

Current technologies

A large number of additive processes are now available. The main differences

between processes are in the way layers are deposited to create parts and in the materials that are used.

Some methods melt or soften the material to produce the layers. Some examples are: selective laser melting (SLM) or direct metal laser sintering (DMLS), selective laser sintering (SLS), fused deposition modeling (FDM) or fused filament fabrication (FFF).

Some other methods cure liquid materials using different sophisticated technologies such as stereolithography (SLA).

"Different types of 3D printers function on a different technology that uses different materials in different ways. One of the most basic limitations of 3D printing in terms of materials and applications is that, there is no one solution that fits all," says Karan Sangani, business development manager, MaherSoft.

Attributes to consider

Selection of the right 3D printer depends on the application requirements and matching the key performance criteria that will provide the best all-around value. Here are

some 3D printer performance attributes to consider when comparing various 3D printers:

Print capacity. Right from prototyping stage, it is critical to print full-size parts to test the product in real conditions, so the build volume (size) is the key. The largest build volume is available with SLA technology.

Speed vs quality. There is a

Fig. 1: A 3D printer and its accessories

